

Textile ERP

Complete Developer Map

Frontend navigation, UI controls, API endpoints, database models, and cross-section data flows

Repository: [textile-erp](#) | Production: <https://erp.kutalimzhda.com>

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1. Application Overview & Architecture

Textile ERP is a full-stack application for a textile business: inventory (rolls, pieces, remanents), point-of-sale, commissions, branch tasks, employee management, and audit history. The React SPA talks to an Express REST API backed by PostgreSQL via Prisma.



2. Technology Stack

Layer	Technology	Location
Frontend	React 19, TypeScript, Vite, React Router, i18next (EN + Kurdish Sorani), Tailwind CSS	frontend/
Backend	Node.js, Express 5, TypeScript (tsx)	src/
ORM / DB	Prisma 6, PostgreSQL	prisma/
Auth	JWT access + refresh tokens, bcrypt passwords	src/lib/auth.ts
Process manager	PM2 (app), cloudflared system service (tunnel)	ecosystem.config.cjs
QR scanning	@zxing/library, camera modal	components/QrScannerModal.tsx

3. Authentication, Roles & Route Guards

Roles

Role	Access Level
ADMIN	Full access; create/edit/delete employees; approve registrations; all sidebar sections
MANAGER	Same route access as ADMIN; cannot create/edit/delete employees (ADMIN-only mutations)
EMPLOYEE	Section-scoped via <code>allowedSections</code> (server) or <code>localStorage</code> rules; always sees Task Employee
TRUSTEE	Same section scoping as EMPLOYEE

Route Guard Chain (`App.tsx` → `ProtectedLayout`)

1. Not authenticated → redirect to `/login`
2. `canAccessAdminRoute` fails → redirect to `/dashboard`
3. `canAccessRoute` fails → redirect to `/task-employee`

Admin-Only Routes (ADMIN or MANAGER to view; some mutations ADMIN-only)

Route	Sidebar Label
<code>/employee-accounts</code>	Employee Accounts
<code>/item-pricing</code>	Item Pricing
<code>/plain-cloth</code>	Plain Cloth Pricing
<code>/sales-commission</code>	Sales Commission

Section Keys (`dashboardSettings.ts`) — used for employee access checkboxes

`dashboard`, `itemPricing`, `salesCommission`, `commissionPayouts`, `inventory`, `itemConversion`, `sales`, `dailySales`, `historySales`, `owedMoney`, `tasks`, `taskEmployee`, `dataAnalysis`, `trusteeCommission`, `exchange`, `itemInput`, `employeeAccounts`, `activityHistory`

Note: `/plain-cloth` is admin-route only but NOT in section keys — cannot be granted via employee checkboxes.

4. Sidebar Navigation (Complete Tree)

File: `frontend/src/components/Sidebar.tsx`. Dropdown groups expand/collapse with ▼ toggle. Footer contains **Language Switcher** (English / Kurdish Sorani).

SIDEBAR		
└─ Dashboard	→ /dashboard	
└─ ▼ Pricing (dropdown)		
└─ Item Pricing	→ /item-pricing	[ADMIN/MANAGER only]
└─ Plain Cloth Pricing	→ /plain-cloth	[ADMIN/MANAGER only]
└─ ▼ Commission (dropdown)		
└─ Sales Commission	→ /sales-commission	[ADMIN/MANAGER only]
└─ Commission Payouts	→ /commission-payouts	
└─ Trustee Commission	→ /trustee-commission	
└─ Inventory	→ /inventory	
└─ Sales	→ /sales	[accent styling]
└─ ▼ Accounting (dropdown)		
└─ Daily Sales	→ /sales/daily	
└─ History Sales	→ /sales/history	
└─ Owed Money	→ /sales/owed	
└─ Exchange	→ /exchange	
└─ Activity History	→ /activity-history	
└─ ▼ Tasks (dropdown)		[red badge if open tasks]
└─ Task Input	→ /tasks	
└─ Task Employee	→ /task-employee	[always visible]
└─ Data Analysis	→ /analytics	
└─ ▼ Item Input (dropdown)		
└─ New Item	→ /item-input	
└─ Item Conversion	→ /inventory/convert	
└─ Employee Accounts	→ /employee-accounts	[ADMIN/MANAGER only]
└─ Language Switcher	(footer, no route)	

Open task badge: Polls `GET /api/tasks/open-count` every 60s. Shows count on Tasks dropdown and Task Employee item. Uses `mine=true` for non-admin/manager users.

5. Global Shell Components

Component	File	Function	API
AppShell	<code>components/AppShell.tsx</code>	Layout wrapper: sidebar + main content; sidebar toggle on mobile	On mount: syncs commission settings via <code>GET/PUT /api/commissions/settings*</code>
Sidebar	<code>components/Sidebar.tsx</code>	Navigation tree, role filtering, task badge	<code>GET /api/tasks/open-count</code>
LanguageSwitcher	<code>components/LanguageSwitcher.tsx</code>	Toggle EN / ckb (Kurdish)	None
QrScanInput	<code>components/QrScanInput.tsx</code>	Text + camera QR input; opens <code>QrScannerModal</code>	Delegated to parent page
QrScannerModal	<code>components/QrScannerModal.tsx</code>	Camera-only QR scanning	None
SignInApprovalsPanel	<code>components/SignInApprovalsPanel.tsx</code>	Pending device sign-in + employee registration approvals	<code>GET /sign-in-requests/pending</code> , <code>GET /notifications/unread-count</code> , <code>approve/reject</code> POSTs

6. Page-by-Page Reference

6.1 Login — /login

File: pages/Login.tsx | **Layout:** Public (no sidebar) | **All users**

UI Controls

- Email input, Password input
- Sign In button
- Link to Register page
- Demo credentials display
- Language Switcher
- Device sign-in pending banner (polls when device awaiting approval)

Backend

Action	Method	Endpoint	Models
Login	POST	/api/auth/login	Employee, Session, DeviceSignInRequest, AuditLog

6.2 Register — /register

File: pages/Register.tsx | Public

UI Controls

- Name, Email, Phone inputs
- Branch **dropdown** (loaded from API)
- Password, Confirm password
- Notes textarea
- Submit button, Language Switcher

Backend

Action	Method	Endpoint
Load branches	GET	/api/auth/branches
Register	POST	/api/auth/register

6.3 Dashboard — /dashboard

File: pages/Dashboard.tsx | **Section key:** dashboard

UI Controls

- Welcome header with user name
- Sign Out button
- SignInApprovalsPanel (device + registration approvals — ADMIN/MANAGER)
- 3 stat cards: role, status, last login
- Overview card

Backend

Action	Method	Endpoint
Sign out	POST	/api/auth/logout
Pending sign-ins	GET	/api/sign-in-requests/pending
Approve/reject device	POST	/api/sign-in-requests/:id/approve reject

Links to: Employee Accounts (via approvals), all sidebar sections

6.4 Item Pricing — /item-pricing

File: pages/ItemPricing.tsx | **ADMIN/MANAGER**

UI Controls

- QR scan input (lookup item by ID)
- Unit **dropdown:** METER / PIECE
- Minimum price input
- Save button
- Scrollable saved-prices list (localStorage cache + server sync)

Backend

Action	Method	Endpoint
Lookup item	GET	/api/inventory/:id
Save minimum price	PUT	/api/commissions/settings/prices

Links to: Inventory (item IDs), Sales Commission (prices affect commission calc)

6.5 Plain Cloth Pricing — /plain-cloth

File: `pages/PlainClothPricing.tsx` | **ADMIN/MANAGER**

UI Controls

- CRUD form: name, price per meter
- Edit / Delete per row
- Offline banner + Reconnect button
- Link back to Sales page

Backend

Action	Method	Endpoint
List types	GET	<code>/api/commissions/plain-cloth</code> or <code>/api/plain-cloth</code>
Create	POST	<code>/api/commissions/plain-cloth</code> or <code>/api/plain-cloth</code>
Update	PUT	<code>/api/commissions/settings/plain-cloth/:id</code> or <code>/api/plain-cloth/:id</code>
Delete	DELETE	same paths

Links to: Sales page (Plain Cloth accordion section uses these types)

6.6 Sales Commission — /sales-commission

File: `pages/SalesCommission.tsx` | **ADMIN/MANAGER**

UI Controls

- Base commission amount input
- Rate % input
- Calculate Commissions button
- Results table (employee, sale, amount)

Backend

Action	Method	Endpoint
Save rate	PUT	<code>/api/commissions/settings/rate</code>
Load sales for calc	GET	<code>/api/sales?pageSize=200</code>

Links to: Commission Payouts, Sales, Item Pricing

6.7 Commission Payouts — /commission-payouts

File: pages/CommissionPayouts.tsx | Section: commissionPayouts

UI Controls

- Grand total display
- Refresh button
- Expandable employee groups → line-level commission details
- Mark Paid button ADMIN/MANAGER only

Backend

Action	Method	Endpoint	Models
List pending	GET	/api/commissions/pending	EmployeeCommissionEntry, Sale, SaleItem
Mark paid	POST	/api/commissions/pay/:employeeId	EmployeeCommissionEntry, PaidCommissionHistory, Notification

6.8 Trustee Commission — /trustee-commission

File: pages/TrusteeCommission.tsx | Section: trusteeCommission

UI Controls

- Date range inputs (from / to)
- Load Commission button
- Rule editor: name, contact, branch toggles (A-F), percentage
- Saved rules list
- 3-level drill-down: Trustee → Branch → Day → individual sales

Backend / Storage

Action	Source
Load sales data	GET /api/sales (per branch, date range)
Trustee rules	localStorage textile-erp-trustee-rules

Schema has Trustee, TrusteePermission models but no backend API yet — rules are client-side only.

6.9 Inventory — /inventory

File: pages/Inventory.tsx | **Section:** inventory

UI Controls

- **QR Search section:** QrScanInput + camera modal → family code lookup
- **Type tabs:** ROLL | PIECE | REMANENT
- **Branch filter buttons:** A, B, C, E, F, S (Storage) — toggle active branch
- **Stock drill-down:** click total → branch breakdown panel
- Piece-package stock subsection
- Items table: checkboxes, reprint QR per row
- Edit/Remove toolbar **ADMIN/MANAGER**
- **Edit modal (overlay):** family code, sub-code price, color select, meters, piece length, quantity

Backend

Action	Method	Endpoint
List items	GET	/api/inventory?branchId=&pageSize=200
QR lookup	GET	/api/inventory/:id
Load colors	GET	/api/inventory/colors
Edit item	PATCH	/api/inventory/:id (requires version for optimistic lock)
Remove (archive)	POST	/api/inventory/:id/archive

Links to: Item Input (creates items), Item Conversion, Sales (scans same QR IDs), Item Pricing

6.10 Item Conversion — `/inventory/convert`

File: `pages/ItemConversion.tsx` | Section: `itemConversion`

UI Controls

- **Section 1 — Branch Transfer:** QR scan, destination branch select, amount input, transfer button
- **Section 2 — Roll to Piece:** QR scan roll, cut meters input, cut button
- Success summary with QR image for new piece

Backend

Action	Method	Endpoint
Lookup source	GET	<code>/api/inventory/:id</code>
Deduct source stock	PATCH	<code>/api/inventory/:id</code>
Create new item	POST	<code>/api/inventory</code>
Complete cutting tasks	POST	<code>/api/tasks/cutting/complete</code>

Links to: Inventory (stock changes), Tasks (auto-completes cutting tasks after roll-to-piece)

6.11 Sales — /sales

File: pages/Sales.tsx | Section: sales

UI Controls

- **Branch selector buttons:** A, B, C, E, F
- **Accordion 1 — Inventory Scan:** QR input, source branch dropdown, amount, price; piece-package mode FULL/PARTIAL with per-component qty
- **Accordion 2 — Roll Cut & Sell:** QR roll scan, cut meters, sell resulting piece
- **Accordion 3 — Plain Cloth:** plain cloth type dropdown, meters, price
- Customer form (name, phone, notes)
- Payment method **dropdown:** CASH | FIB | OWED
- Partial payment fields (when OWED or partial)
- Sale lines cart with line totals
- Complete Sale button

Backend

Action	Method	Endpoint	Side Effects
Lookup inventory	GET	/api/inventory/:id , /api/inventory	—
Cut roll to piece	PATCH+POST	/api/inventory/:id , /api/inventory	Creates piece, deducts roll
Complete sale	POST	/api/sales	Deducts stock, creates Sale+SaleItems, EmployeeCommissionEntry, may create BranchTaskEntry (cutting)
Plain cloth types	GET	/api/commissions/plain-cloth	—
Cutting task on piece sold	POST	/api/tasks	When piece sold from roll
Complete cutting tasks	POST	/api/tasks/cutting/complete	After roll cut

Links to: Inventory, Plain Cloth Pricing, Commission Payouts, Daily/History Sales, Tasks

6.12 Exchange — /exchange

File: pages/Exchange.tsx | Section: exchange

UI Controls

- Branch selector buttons
- New Exchange Sale section (replacement items via QR)
- Returned Inventory section (QR scan returns)
- Returned Plain Cloth **dropdown**
- Exchange Summary totals
- Customer Details form
- Payment: FULL / PARTIAL
- Process Exchange button

Backend

Action	Method	Endpoint
Process exchange	POST	/api/sales/exchange
Resolve inventory items	GET	/api/inventory/:id, /api/inventory
Plain cloth types	GET	/api/commissions/plain-cloth

6.13 New Item (Item Input) — `/item-input`

File: `pages/ItemInput.tsx` | **Section:** `itemInput`

UI Controls

- Destination branch buttons (A, B, C, E, F, S)
- Type selector: ROLL / PIECE / REMANENT
- Color **dropdown** + add-new-color form **ADMIN/MANAGER**
- Family code, sub-code inputs
- Meters, quantity, piece length inputs
- Piece-package component builder (multi-component packages)
- Picture upload, description textarea
- Live QR preview
- Create button → post-create Print / Download QR actions

Backend

Action	Method	Endpoint
List colors	GET	<code>/api/inventory/colors</code>
Create color	POST	<code>/api/inventory/colors</code>
Family lookup	GET	<code>/api/inventory</code>
Create item	POST	<code>/api/inventory</code>

Links to: Inventory (new items appear in list), Sales (QR scannable immediately)

6.14 Sale Detail — /sales/:id (drill-down, not in sidebar)

File: pages/SaleDetail.tsx

Reached from

Daily Sales, History Sales, Owed Money, Activity History (related record link)

UI Controls

- Back button
- Sale metadata grid (ID, date, employee, branch, customer)
- Price / discount / payment method cards
- Partial payment breakdown
- Notes section
- Line items table with QR download per line

Backend

GET	/api/sales/:id	Sale, SaleItem, Refund, InventoryItem, Color
-----	----------------	--

6.15 Daily Sales — /sales/daily

File: pages/DailySales.tsx | Section: dailySales

UI Controls

- Branch filter buttons
- Sales grouped by employee
- Clickable sale rows → drill to /sales/:id

Backend

GET /api/sales?branchId=&fromDate=&toDate= (today). Owed payments merged from localStorage .

6.16 History Sales — /sales/history

File: `pages/HistorySales.tsx` | **Section:** `historySales`

UI Controls

- Branch filter buttons
- **Date tabs:** last 10 days (toggle)
- Search input + Apply button
- Employee-grouped sales → drill to `/sales/:id`

Backend

`GET /api/sales` with date buckets; search uses `search` + `pageSize=100`

6.17 Owed Money — /sales/owed

File: `pages/OwedMoney.tsx` | **Section:** `owedMoney`

UI Controls

- Branch filter buttons
- Outstanding / settled sale rows
- Record Payment button → **payment modal (overlay)**
- Navigate to sale detail

Backend

`GET /api/sales?branchId=&includeVoided=false&pageSize=200`

Important: Payment recordings stored in `localStorage` `textile-erp-owed-payments` only — NOT persisted to backend.

6.18 Task Input — /tasks

File: pages/Tasks.tsx | Section: tasks

UI Controls

- Branch filter buttons
- Assign recurring form: task template **dropdown**, schedule **dropdown**, email, notes
- Manual cutting task (prompt dialog)
- Stats cards (open / done counts)
- Assignment list
- Open tasks list: Done / Reopen / Delete / Create Next buttons
- Done tasks list

Backend (lib/tasksApi.ts)

Action	Method	Endpoint
List tasks	GET	/api/tasks?branch=
List assignments	GET	/api/tasks/assignments?branch=
Upsert assignment	PUT	/api/tasks/assignments
Create task	POST	/api/tasks
Complete	PATCH	/api/tasks/:id/complete
Reopen	PATCH	/api/tasks/:id/reopen
Delete	DELETE	/api/tasks/:id

Links to: Sales & Item Conversion (cutting tasks auto-created/completed), Task Employee

6.19 Task Employee — /task-employee

File: pages/TaskEmployee.tsx | Always visible (fallback route)

UI Controls

- Branch filter buttons
- Open / done stats
- Checklist with Mark Done per task

Backend

GET /api/tasks?branch=&mine=true , PATCH /api/tasks/:id/complete

6.20 Data Analysis — /analytics

File: pages/DataAnalysis.tsx | **Section:** dataAnalysis

UI Controls

- Branch **dropdown**
- From / To date inputs
- Load Analysis button
- Summary cards (revenue, total sales, average sale)
- Bar charts: sales by day, sales by employee

Backend

GET /api/sales?branchId=&fromDate=&toDate=&includeVoided=false&pageSize=200

6.21 Activity History — /activity-history

File: pages/ActivityHistory.tsx | **Section:** activityHistory

UI Controls

- From / To date inputs
- Action **dropdown** (CREATE, UPDATE, DELETE, VOID, LOGIN...)
- Entity type **dropdown**
- Search input + Apply Filters
- Date-grouped clickable entries → **drill to** /activity-history/:id
- Load More pagination

Backend (lib/auditLogApi.ts)

GET /api/audit-logs (paginated), GET /api/audit-logs/entity-types

6.22 Activity History Detail — /activity-history/:id

File: pages/ActivityHistoryDetail.tsx

UI Controls

- Back button
- Performed-by section
- Action info, changes JSON diff
- Related record snapshot + optional link to Sale/Inventory

Backend

GET /api/audit-logs/:id

6.23 Employee Accounts — /employee-accounts

File: `pages/EmployeeAccounts.tsx` | **ADMIN/MANAGER view** , **ADMIN mutate**

UI Controls

- `SignInApprovalsPanel`
- Create/Edit form **ADMIN only** : name, email, phone, password, role select, active checkbox, assigned work, branch checkboxes, sidebar-access checkboxes (all `dashboardSections`)
- Employee table with Edit / Delete **ADMIN only**

Backend

Action	Method	Endpoint	Auth
List employees	GET	<code>/api/employees</code>	ADMIN/MANAGER
List branches	GET	<code>/api/employees/branches/list</code>	ADMIN/MANAGER
Create	POST	<code>/api/employees</code>	ADMIN
Update	PATCH	<code>/api/employees/:id</code>	ADMIN
Delete	DELETE	<code>/api/employees/:id</code>	ADMIN

Links to: All sections (`controls allowedSections` per employee), Dashboard (sign-in approvals)

7. Drill-Down & Cross-Navigation Map

From	User Action	To
Daily Sales	Click sale row	/sales/:id (Sale Detail)
History Sales	Click sale row	/sales/:id
Owed Money	Click sale row	/sales/:id
Activity History	Click log entry	/activity-history/:id
Activity History Detail	Related record link	/sales/:id or inventory context
Inventory	Click stock total	Branch breakdown panel (in-page drill-down)
Trustee Commission	Click trustee → branch → day	3-level in-page drill to individual sales
Commission Payouts	Expand employee group	Line-level commission details (in-page)
Plain Cloth Pricing	"Back to Sales" link	/sales
Login	Register link	/register
Any protected route	Access denied	/task-employee or /dashboard

8. Backend API Reference (All 71 Endpoints)

8.1 Server-Level

Method	Path	Auth	Purpose
GET	/health	None	DB connectivity check
GET	/api/version	None	Deploy metadata
GET	/	None	SPA index (production)

8.2 Auth — /api/auth (auth.routes.ts)

Method	Path	Auth	Models
GET	/branches	None	Branch
POST	/register	None	Employee, BranchEmployee, AuditLog
POST	/login	None	Employee, Session, DeviceSignInRequest
POST	/refresh	None	Session, Employee
POST	/logout	None	Session
GET	/me	JWT	Employee, BranchEmployee, Branch

8.3 Inventory — `/api/inventory` (all routes require JWT)

Method	Path	Extra Auth	Purpose
GET	<code>/</code>	Any	Paginated list (filters: branchId, colorId, type, code, itemId)
GET	<code>/stats/summary</code>	Any	Inventory stats by branch
GET	<code>/branches</code>	Any	Active branches
GET	<code>/colors</code>	Any	Active colors
POST	<code>/colors</code>	ADMIN/MANAGER	Create color
GET	<code>/:id</code>	Any	Single item
POST	<code>/</code>	ADMIN/MANAGER	Create item
PATCH	<code>/:id</code>	ADMIN/MANAGER	Update (optimistic lock via version)
POST	<code>/:id/archive</code>	ADMIN/MANAGER	Soft-archive
POST	<code>/:id/restore</code>	ADMIN/MANAGER	Restore archived
DELETE	<code>/:id</code>	ADMIN	Permanent delete

8.4 Sales — `/api/sales`

Method	Path	Extra Auth	Side Effects
POST	<code>/</code>	Any	Create sale, deduct stock, commission entries, notifications
POST	<code>/exchange</code>	Any	Exchange: returns + replacements, stock adjustments
GET	<code>/</code>	Any	List with filters (branchId, dates, search, pagination)
GET	<code>/stats/summary</code>	ADMIN/MANAGER	Sales aggregates
GET	<code>/:id</code>	Any	Sale detail with items
POST	<code>/:id/void</code>	ADMIN/MANAGER	Void sale, restock, remove pending commissions
POST	<code>/:id/refund</code>	ADMIN/MANAGER	Record refund

8.5 Employees — `/api/employees` (router: ADMIN/MANAGER)

Method	Path	Extra Auth
GET	<code>/</code>	ADMIN/MANAGER
GET	<code>/sections</code>	ADMIN/MANAGER
GET	<code>/branches/list</code>	ADMIN/MANAGER
GET	<code>/pending</code>	ADMIN/MANAGER
POST	<code>/:id/approve</code>	ADMIN
POST	<code>/:id/reject</code>	ADMIN
GET	<code>/:id</code>	ADMIN/MANAGER
POST	<code>/</code>	ADMIN
PATCH	<code>/:id</code>	ADMIN
DELETE	<code>/:id</code>	ADMIN

8.6 Commissions — `/api/commissions`

Method	Path	Extra Auth
GET/POST/PUT/DELETE	<code>/plain-cloth/*</code>	Varies
GET	<code>/settings</code>	Any
PUT	<code>/settings/rate</code>	ADMIN/MANAGER
PUT	<code>/settings/prices</code>	ADMIN/MANAGER
GET	<code>/pending</code>	Any (own only for non-managers)
POST	<code>/pay/:employeeId</code>	ADMIN/MANAGER
POST	<code>/backfill</code>	ADMIN

8.7 Tasks — `/api/tasks`

Method	Path	Extra Auth	Model
GET	<code>/open-count</code>	Any	BranchTaskEntry
GET/PUT	<code>/assignments</code>	ADMIN/MANAGER	BranchTaskAssignment
GET	<code>/cutting/open</code>	Any	BranchTaskEntry
POST	<code>/cutting/complete</code>	All roles	BranchTaskEntry
GET/POST	<code>/</code>	Any	BranchTaskEntry
PATCH	<code>/:id/complete reopen</code>	Any / ADMIN/MANAGER	BranchTaskEntry
DELETE	<code>/:id</code>	ADMIN/MANAGER	BranchTaskEntry

8.8 Other Modules

Prefix	Endpoints	Key Models
/api/notifications	GET /, GET /unread-count, PATCH /read-all, PATCH /:id/read	Notification
/api/sign-in-requests	GET /pending, POST /:id/approve, POST /:id/reject	DeviceSignInRequest
/api/audit-logs	GET /, GET /entity-types, GET /:id	AuditLog
/api/plain-cloth	GET/POST/PUT/DELETE (with audit logs on mutations)	PlainClothPricing

9. Database Schema & Entity Relationships

Core Business Entities

Model	Table	Key Fields	Related To
Branch	branches	id (A,B,C...), name	Inventory, Sales, Tasks, Employees
InventoryItem	inventory_items	id (QR), code, subCode, type, meters, quantity	Branch, Color, SaleItem
Color	colors	name, hexCode	InventoryItem, SaleItem
Sale	sales	branchId, employeeId, customerName, paymentMethod, total	SaleItem, Refund, EmployeeCommissionEntry
SaleItem	sale_items	inventoryItemId, soldUnit, quantity, unitPrice	Sale, InventoryItem, Color
Employee	employees	email, role, allowedSections, approvalStatus	Session, BranchEmployee, Sales
EmployeeCommissionEntry	employee_commission_entries	status PENDING/PAID, amount	Sale, SaleItem, Employee
BranchTaskEntry	branch_task_entries	templateKey, assigneeId, doneAt	Branch, Employee
PlainClothPricing	plain_cloth_pricing	name, pricePerMeter	Used in sales (non-inventory lines)
AuditLog	audit_logs	action, entityType, entityId, changes	Employee, Branch

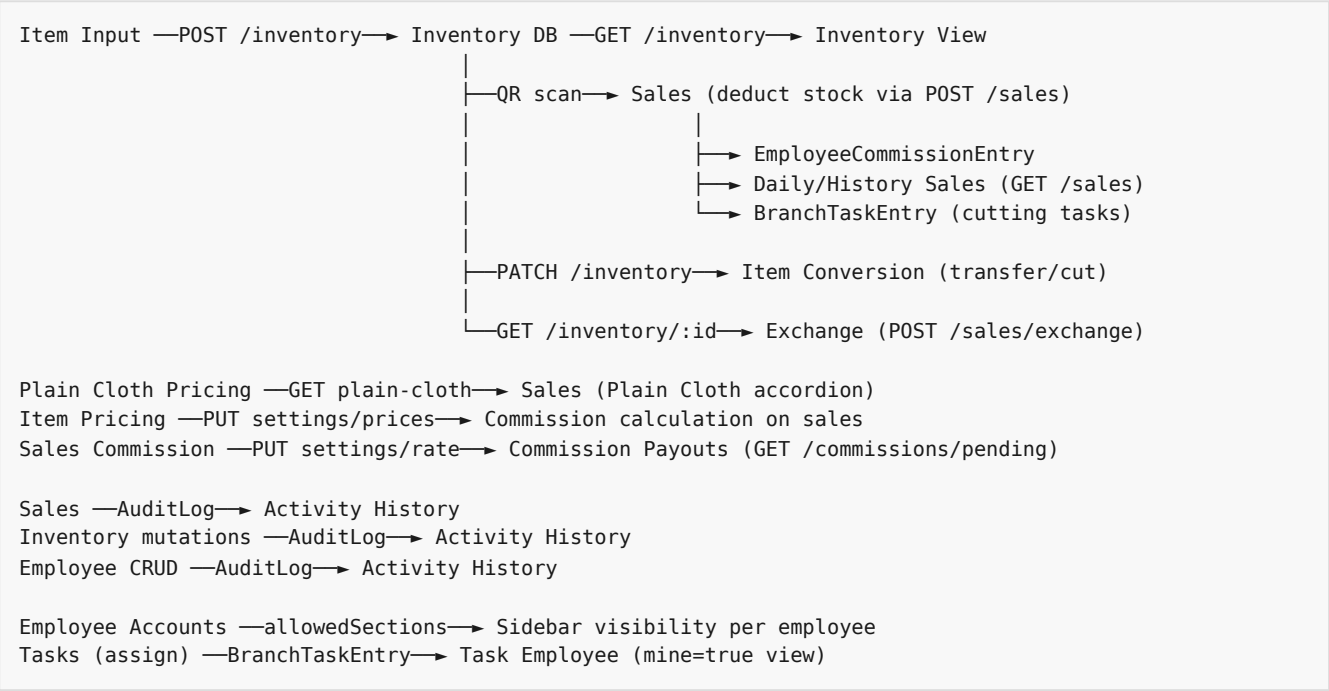
Inventory Item ID Format (QR)

Composite ID encoding branch + family code + sub-code + color + type + piece length. Parsed by `lib/inventoryCodes.ts` . Scanned on Sales, Inventory, Exchange, Item Conversion, Item Pricing, Item Input.

Branches

Codes: **A, B, C, E, F** (shop floors) and **S** (Storage). Defined in `lib/inventoryCodes.ts` as `BRANCH_DESTINATIONS` .

10. Section-to-Section Data Flow



11. Client-Only vs Server-Backed Features

Feature	Storage	Pages Affected
Owed payment recordings	localStorage textile-erp-owed-payments	Owed Money, Daily Sales
Trustee commission rules	localStorage textile-erp-trustee-rules	Trustee Commission
Item minimum prices cache	localStorage + server sync	Item Pricing
Commission settings cache	localStorage + server sync	Sales Commission, AppShell
Employee access rules (fallback)	localStorage textile-erp-employee-access-rules	Route guards (if server sections empty)
JWT tokens + user profile	localStorage	Auth, all API calls

12. Shared Libraries & Key Files

File	Purpose
frontend/src/lib/api.ts	Axios client, JWT interceptor, retry on 530/502/503
frontend/src/lib/auth.ts	Login/logout, getCurrentUser, role helpers
frontend/src/lib/dashboardSettings.ts	Section keys, route access, localStorage settings
frontend/src/lib/inventoryCodes.ts	QR parsing, branch codes, stock aggregation
frontend/src/lib/salesFlow.ts	Roll cut, piece sell, cutting task hooks
frontend/src/lib/tasksApi.ts	All task API wrappers
frontend/src/lib/auditLogApi.ts	Activity history API wrappers
frontend/src/lib/plainClothApi.ts	Plain cloth CRUD with fallback paths
src/server.ts	Express entry, route mounting, SPA serve
src/middleware/authenticate.ts	JWT verify, requireRole
src/lib/inventory.ts	Inventory business logic
src/lib/sales.ts	Sales + exchange logic
src/lib/commissions.ts	Commission calculation + payout
src/lib/branchTasks.ts	Branch task CRUD
prisma/schema.prisma	Full database schema (22 models)

Textile ERP Developer Map — End of Document
For updates, regenerate from repository main branch.